In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.datasets import cifar10, fashion_mnist

(cifar_x, _), _ = cifar10.load_data()
(fashion_x, _), _ = fashion_mnist.load_data()

print("Task 1: Computation")
print("CIFAR-10 Original: 32 x 32 -> Resized: 16 x 16")
print("Fashion-MNIST Original: 28 x 28 -> Resized: 14 x 14")

cifar_resized = []
fashion_resized = []

for i in range(5):
    img = tf.image.resize(cifar_x[i], (16, 16)).numpy().astype(np.uint8)
    cifar_resized.append(img)

for i in range(5):
    img = tf.image.resize(
        fashion_x[i].reshape(28, 28, 1),
        (14, 14)
    ).numpy().squeeze()
    fashion_resized.append(img)

plt.figure(figsize=(15,6))

for i in range(5):
    plt.subplot(2,5,i+1)
    plt.imshow(cifar_resized[i])
    plt.title("Resized (16x16)")
    plt.axis("off")

for i in range(5):
    plt.subplot(2,5,i+6)
    plt.imshow(fashion_x[i], cmap="gray")
    plt.title("Original (28x28)")
    plt.axis("off")

plt.tight_layout()
plt.show()

plt.figure(figsize=(15,6))

for i in range(5):
    plt.subplot(2,5,i+1)
    plt.imshow(cifar_x[i])
    plt.title("Original (32x32)")
    plt.axis("off")

for i in range(5):
    plt.subplot(2,5,i+6)
    plt.imshow(fashion_resized[i], cmap="gray")
    plt.title("Resized (14x14)")
    plt.axis("off")
```

```python
plt.tight_layout()
plt.show()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/train-labels-idx1-ubyte.gz
29515/29515 ━━━━━━━━━━━━━━━━━━━━ 0s 3us/step
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26421880/26421880 ━━━━━━━━━━━━━━━━━━━━ 15s 1us/step
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Task 1: Computation
CIFAR-10 Original: 32 x 32 -> Resized: 16 x 16
Fashion-MNIST Original: 28 x 28 -> Resized: 14 x 14
```

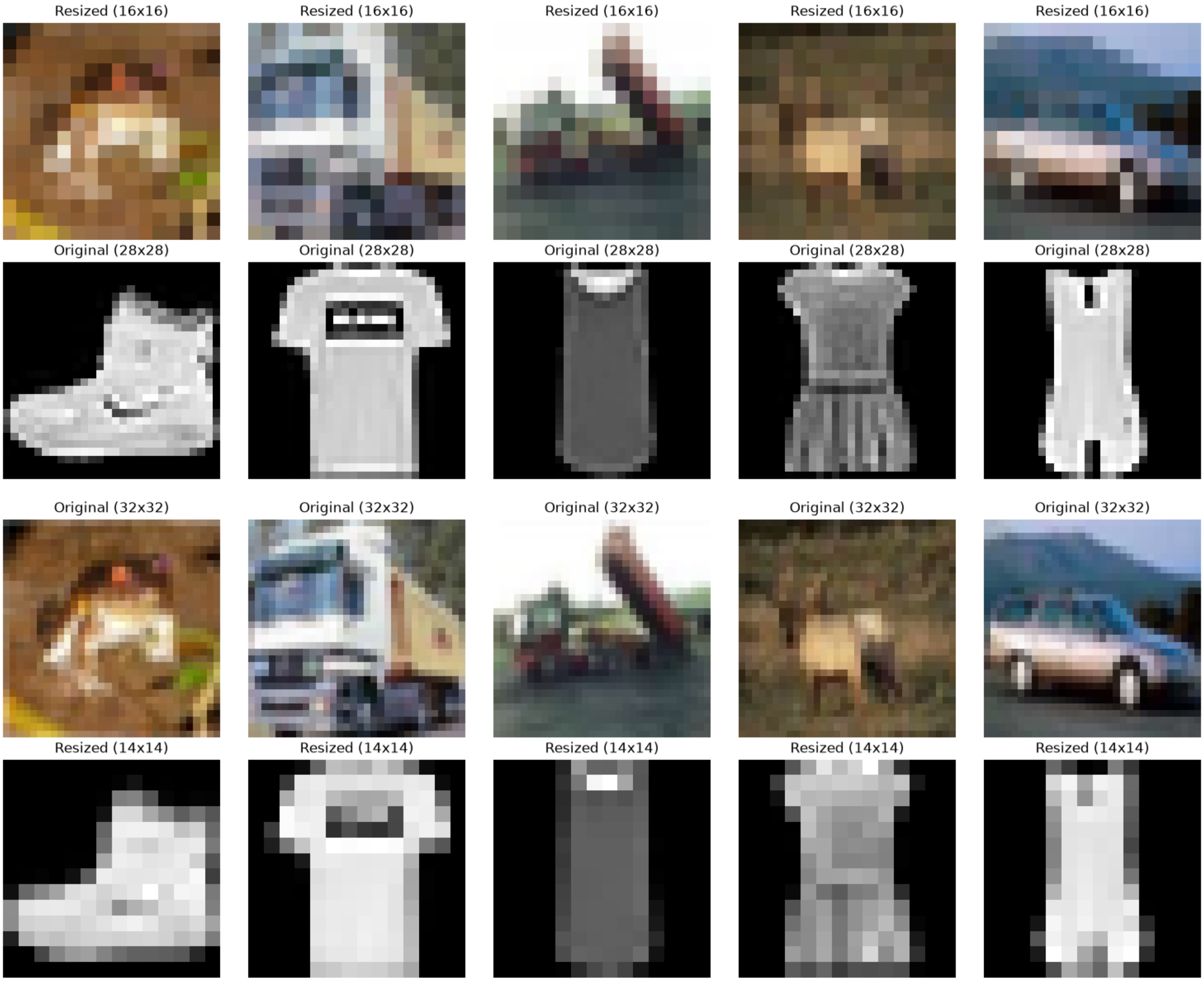

In [ ]: